

Marks

2

3

3

3

$$\frac{4x-6}{(x+1)(2x^2+3)} \equiv \frac{a}{x+1} + \frac{bx+c}{2x^2+3} \quad \text{where } a, b \text{ and } c \text{ are real numbers}$$

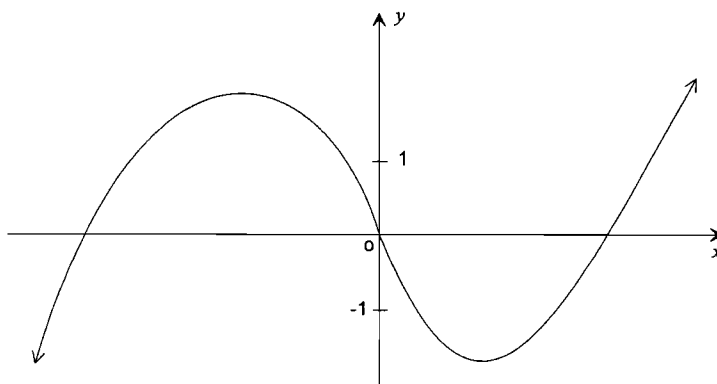
2

2

QUESTION 2 (15 marks) Start on a new page

Marks

(a) The diagram shows the graph of $y = f(x)$



Draw separate sketches (at least $\frac{1}{3}$ of a page) of the graphs of the following

i) $y = \frac{1}{f(x)}$ 2

ii) $y = f(|x|)$ 2

iii) $y = \log_e f(x)$ 2

iv) $y^2 = f(x)$ 2

(b) Let $f(x) = \frac{x-2}{(x+1)(5-x)}$ 3

Draw a neat sketch of $y = f(x)$ showing all asymptotes and intercepts with the coordinate axes. You are not required to use calculus.

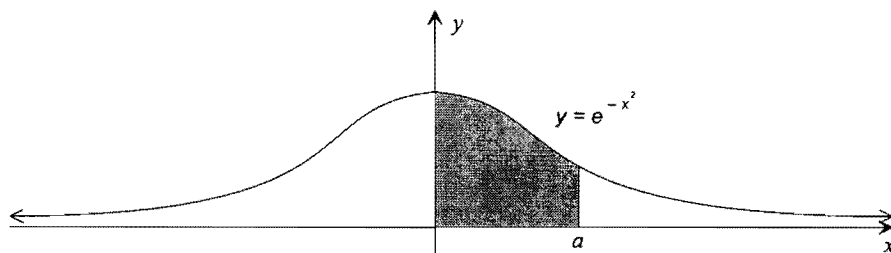
(c) If $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$. prove that

$$I_n = \frac{n-1}{n} I_{n-2} \quad \text{and hence find} \quad 3$$

$$\int_0^{\frac{\pi}{2}} \cos^5 x \, dx. \quad 1$$

- (a) The region bounded by the curve $y = e^{-x^2}$, the coordinate axes and the line $x = a$ ($a > 0$) is rotated about the y axis to form a solid of revolution.

- i) Use the method of cylindrical shells to find the volume of the solid. 4
- ii) What is the limiting value of the volume as $a \rightarrow \infty$. 1



- (b) The base of a solid is the region in the xy plane enclosed by the ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

Each cross section perpendicular to the x axis is a semi ellipse whose major axis is in the base and whose minor axis is half the major axis.

- i) Find the volume V_1 of the solid given that the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $A = \pi ab$ 4
- ii) If the base of this solid was rotated about the x axis to form a solid of volume V_2 find the ratio $V_1 : V_2$ 1

- (c) Use result that

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx \text{ to show that}$$

$$\int_0^{\frac{\pi}{4}} \frac{\operatorname{cosec} 2x - 1}{\operatorname{cosec} 2x + 1} dx = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2x}{1 + \cos 2x} dx \quad 3$$

Hence evaluate $\int_0^{\frac{\pi}{4}} \frac{\operatorname{cosec} 2x - 1}{\operatorname{cosec} 2x + 1} dx$ 2

END OF EXAMINATION

$$1. a) i) \int_1^6 \frac{dx}{\sqrt{x+3}}$$

$$= \left[2\sqrt{x+3} \right]_1^6$$

$$= 2\sqrt{9} - 2\sqrt{4}$$

$$= 2$$

$$ii) \int \frac{dx}{3+2\cos x}$$

$$\text{let } t = \tan \frac{x}{2}$$

$$\therefore I = \int \frac{1}{3 + \frac{2(1-t^2)}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$= \int \frac{2dt}{5+t^2}$$

$$= \frac{2}{\sqrt{5}} \cdot \tan^{-1} \frac{t}{\sqrt{5}}$$

$$= \frac{2}{\sqrt{5}} \tan^{-1} \left(\frac{\tan \frac{x}{2}}{\sqrt{5}} \right) + C$$

$$iii) \int \sqrt{\frac{2-x}{3+x}} dx$$

$$= \int \frac{2-x}{\sqrt{(3+x)(2-x)}} dx$$

$$= \int \frac{2-x}{\sqrt{6-x-x^2}} dx$$

$$= \int \frac{\frac{1}{2}(1-x) + \frac{5}{2}}{\sqrt{6-x-x^2}} dx$$

$$= \sqrt{6-x-x^2} + \frac{5}{2} \int \frac{dx}{\sqrt{6-(x^2+x+\frac{1}{4})+\frac{1}{4}}}$$

$$= \sqrt{6-x-x^2} + \frac{5}{2} \int \frac{du}{\sqrt{\frac{45}{4} - (u+\frac{1}{2})^2}}$$

$$= \sqrt{6-x-x^2} + \frac{5}{2} \ln^{-1} \left(\frac{u+\frac{1}{2}}{\frac{\sqrt{45}}{2}} \right)$$

$$= \sqrt{6-x-x^2} + \frac{5}{2} \ln^{-1} \left(\frac{2x+1}{\sqrt{45}} \right) + C$$

$$iv) \int \sqrt{x} \ln x dx$$

$$\text{let } u = \ln x \quad v' = x^{\frac{1}{2}}$$

$$u' = \frac{1}{x} \quad v = \frac{2}{3} x^{\frac{3}{2}}$$

$$\therefore I = \frac{2}{3} x \sqrt{x} \ln x - \frac{2}{3} \int x^{\frac{1}{2}} dx$$

$$= \frac{2}{3} x \sqrt{x} \ln x - \frac{4}{9} x \sqrt{x} + C$$

$$b) \frac{4x-6}{(x+1)(2x^2+3)} \equiv \frac{a}{x+1} + \frac{bx+c}{2x^2+3}$$

$$a(2x^2+3) + (bx+c)(x+1) \equiv 4x-6$$

$$\text{SUBST } x = -1 \Rightarrow 5a = -10 \therefore a = -2$$

$$\text{COEF OF } x^2 \Rightarrow 2a + b = 0$$

$$\therefore -4 + b = 0 \therefore b = 4$$

$$\text{CONST TERM} \Rightarrow 3a + c = -6$$

$$-6 + c = -6 \therefore c = 0$$

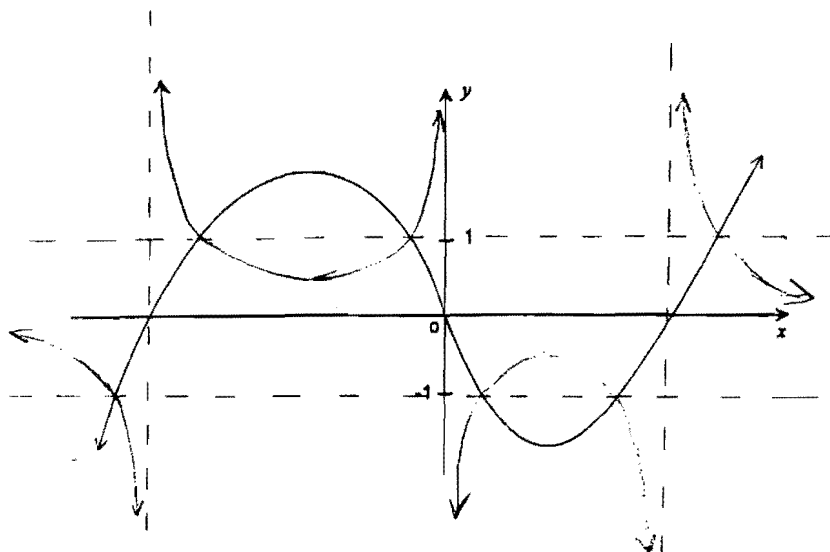
$$\therefore I = \int \frac{4x dx}{2x^2+3} - \int \frac{2}{x+1} dx$$

$$= \ln(2x^2+3) - 2 \ln|x+1| + C$$

$$= \ln \left(\frac{2x^2+3}{(x+1)^2} \right) + C$$

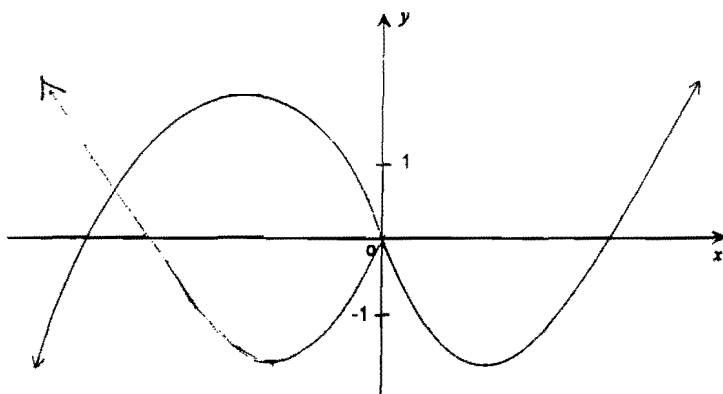
2 a)

i)



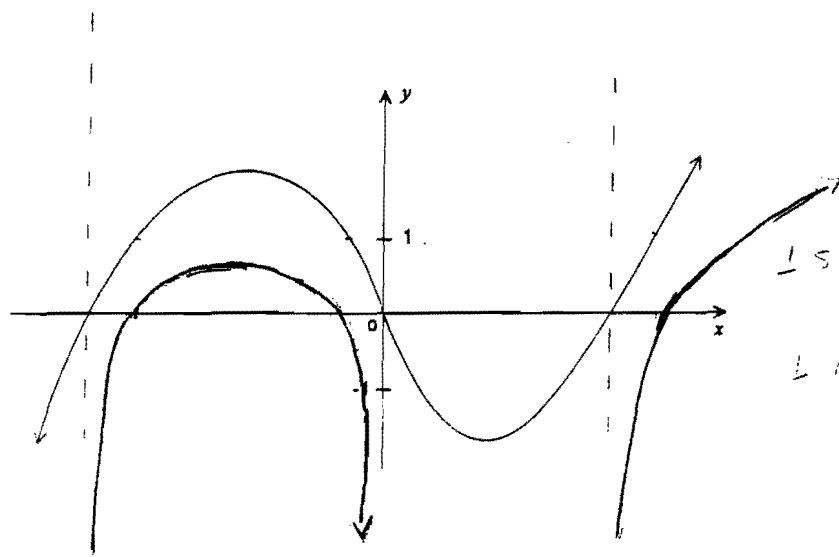
1 GENERAL SHAPE (2)
1 ASYMPTOTES AND DISCONTINUITIES

ii)



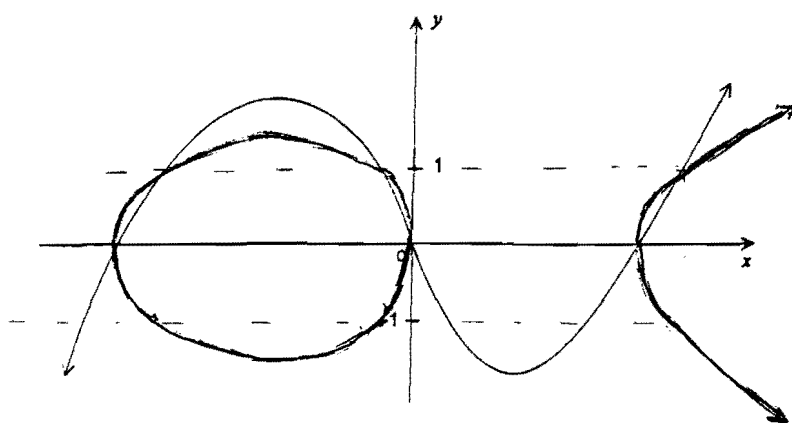
1 LEFT HAND (2)
1 RIGHT HAND

iii)



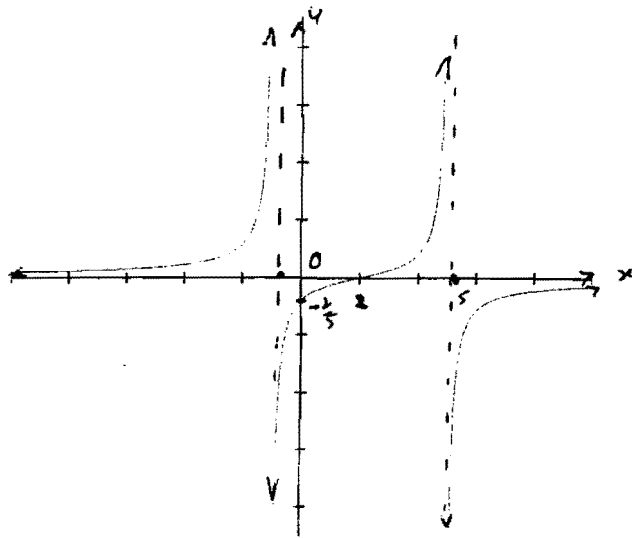
1 SHAPE INCLUDING DISCONTINUITIES (2)
1 INTERCEPTS

iv)



1 SHAPE WRT. DISCONTINUITIES (2)
1 VERTICAL OF HORIZONTAL PTS WHERE $y = \pm 1$

b)



1 GENERAL SHAPE

1 ASYMPTOTES

1 X, Y INTERCEPTS

(3)

$$c) I_n = \int_0^{\pi/2} \cos^n x dx$$

$$\int_0^{\pi/2} \cos^5 x dx = I_5$$

$$\text{Let } u = \cos^{n-1} x \quad u' = \cos x$$

$$u' = (n-1) \cos^{n-2} x \sin x \quad v = \sin x \quad \perp$$

$$I_5 = \frac{4}{5} I_3$$

$$\therefore I_n = \left[\cos^{n-1} x \sin x \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \cos^{n-2} x \sin^2 x dx$$

$$I_3 = \frac{2}{3} I_1$$

$$= 0 + (n-1) \int_0^{\pi/2} (\cos^{n-2} x - \cos^n x) dx$$

$$I_1 = \int_0^{\pi/2} \cos x dx$$

$$= \left[\sin x \right]_0^{\pi/2} = 1 \quad (1)$$

$$= (n-1) \int_0^{\pi/2} \cos^{n-2} x dx - (n-1) \int_0^{\pi/2} \cos^n x dx$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$\therefore I_5 = \frac{4}{5} \cdot \frac{2}{3} \cdot 1$$

$$(n-1+1) I_n = (n-1) I_{n-2} \quad (3)$$

$$= \frac{8}{15} \quad \perp$$

$$I_n = \frac{n-1}{n} I_{n-2} \quad \perp$$

15

③ a) i) $V = 2\pi \int_0^a xy \, dy$ ii) for V_2 $SA = \pi \cdot y^2$

$$= 2\pi \int_0^a x e^{-x^2} \, dx$$

$$= 2\pi \left[-\frac{1}{2} e^{-x^2} \right]_0^a$$

$$= 2\pi \left[-\frac{1}{2} e^{-a^2} + \frac{1}{2} \right]$$

$$= \pi \left[1 - e^{-a^2} \right] \text{ unit}^2$$

ii) as $a \rightarrow \infty$

$$e^{-a^2} \rightarrow 0$$

$\therefore$ LIMITING VALUE OF

$$V = \pi \text{ unit}^2$$

b) i)

$$V = 2 \int_0^4 \pi \frac{y^2}{4} \, dx$$

$$= \frac{\pi}{2} \int_0^4 y^2 \, dx$$

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

$$\therefore y^2 = 9 \left(1 - \frac{x^2}{16} \right)$$

$$\therefore V = \frac{9\pi}{2} \int_0^4 \left(1 - \frac{x^2}{16} \right) \, dx$$

$$= \frac{9\pi}{2} \left[x - \frac{x^3}{48} \right]_0^4$$

$$= \frac{9\pi}{2} \left[4 - \frac{4}{3} \right]$$

$$= 12\pi \text{ unit}^2$$

$$\therefore V_2 = 4V_1$$

$$\therefore V_1 : V_2 = 1 : 4$$

c) $I = \int_0^{\frac{\pi}{4}} \frac{\sec 2\left(\frac{\pi}{4} - x\right) - 1}{\sec 2\left(\frac{\pi}{4} - x\right) + 1} \, dx$

$$= \int_0^{\frac{\pi}{4}} \frac{\sec\left(\frac{\pi}{2} - 2x\right) - 1}{\sec\left(\frac{\pi}{2} - 2x\right) + 1} \, dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\frac{1}{\cos} - 1}{\frac{1}{\cos} + 1} \, dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2x}{1 + \cos 2x} \, dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{1 - (1 - 2\sin^2 x)}{1 + (2\cos^2 x - 1)} \, dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{2\sin^2 x}{2\cos^2 x} \, dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan^2 x - 1) \, dx$$

$$= \left[\tan x - x \right]_0^{\frac{\pi}{4}}$$

$$= 1 - \frac{\pi}{4}$$

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